F78AB - Actuarial and Financial Mathematics B
Tutorial 8 – Arbitrage and forward contracts

Question 1 An investor is able to purchase or sell two specially designed risk-free securities, A and B. Short sales of both securities are possible. Security A has a market price of 20p. In the event that a particular stock market index goes up over the next year, it will pay 25p and, in the event that the stock market index goes down, it will pay 15p. Security B has a market price of 15p. In the event that the stock market index goes up over the next year, it will pay 20p and, in the event that the stock market index goes down, it will pay 12p.

1. Explain what is meant by the assumption of no-arbitrage in a financial market.
2. Find the no-arbitrage price of B, assuming that the market price of A is fixed.
3. Find an arbitrage strategy based on the market price of B being 15p.

Question 2 Two assets A and B are traded over one time period on a stock exchange. Suppose that during this period market prices either go up or down.

		Price at time 1	
Asset	Price at time 0	Market falls	Market rises
A	10	5	15
B	6	4	12

Show how you can make an arbitrage profit by investing in these two assets.

Question 3 Risky asset A has a current price of £100. In one year, asset A will have a price of £90 if the market goes down and a price of £125 if the market goes up.

Risky asset B has a current price of P_B. In one year, asset B will have a price of £40 if the market goes down and a price of £55 if the market goes up.

Investors can borrow or deposit money from a bank account at a rate of 5% per annum effective.

1. Derive the no-arbitrage price P_B^{fair} for asset B, i.e. the price that will not create any arbitrage opportunities in the market.

   (Hint: replicate the possible payoffs on asset B using the bank acccount and asset A.)
2. If the market price P_B for asset B is less than its no-arbitrage price P_B^{fair}, show how an arbitrage profit can be made.

Question 4 Consider a forward contract to be set up at time $t = 0$ on a non-dividend-paying stock that has price S_t at time t. Assume $S_0 = 80$, the delivery date is $T = 4$ months and the risk-free force of interest is 6% per annum.

1. Calculate the no-arbitrage delivery price for this forward contract.
2. Explain why the no-arbitrage delivery price does not depend on the statistical model for S_t.

Question 5 Consider a forward contract to be set up at time $t = 0$ on a non-dividend-paying stock that has price S_t at time t. Assume $S_0 = 100$, the delivery date is $T = 6$ months, the risk-free force of interest is 5% per annum and the forward price is $K = 102$.

Detail an investment strategy that will deliver an arbitrage profit in 6 months' time.

Question 6 The current price of a dividend-paying stock is £10 000 and the risk free force of interest is 5% p.a. A dividend of £600 will be paid in 3 months' time.

1. Calculate the no-arbitrage forward price for this stock, for a delivery time in 9 months.

Suppose that a forward contract is struck on the stock with delivery time in 9 months and forward price £10 450.

2. Explain how you can make an arbitrage profit, using an appropriate combination of the stock and a bank account which gives on deposits and charges on loans the risk-free force of interest.

Question 7 A company based in the UK knows that it will receive a payment of USD10 million in 3 months' time. The company wishes to eliminate its exposure to foreign exchange (FX) risk by entering into a forward contract. Today,

- GBP1 = USD1.55432;
- in the UK, the 3-month interest rate is 1.7% per annum effective; and
- in the USA, the 3-month interest rate is 0.75% per annum effective.

Derive the complete terms, including any numerical values, of the forward contract.

Question 8 Consider a share index that pays dividends. Denote the value of a unit of the index by S_t at time t. One unit of the index provides a known annual dividend yield $g > 0$, i.e. cash dividends are payable continuously at the rate $\rho(t) = gS_t$ per unit of the index at time t.

A forward contract has been agreed to pay the delivery price K at the delivery date T in return for one unit of the index. The annualised risk-free force of interest is δ.

Use a replication argument to show that the no-arbitrage delivery price that ensures the contract has zero value at time 0 is

$$K = S_0 e^{(\delta - g)T}.$$

Question 9 Consider a share index that pays dividends and has value S_t at time t per unit of the index. Assume that $S_0 = 1\,000$ and that one unit of the index provides a known dividend yield with cash dividends payable continuously at the rate

$$\rho(t) = \begin{cases} 0.05S_t & \text{if } 0 \leq t < 1 \\ 0.03S_t & \text{if } 1 \leq t < 2. \end{cases}$$

A forward contract has been arranged to pay the delivery price K at the delivery date $T = 2$ in return for one unit of the index. The annualised risk-free force of interest is $\delta = 0.045$.

Calculate the no-arbitrage forward price for this contract.

Question 10 The current price of a dividend-paying stock is £100. Denote by S_t the price at time t of the stock immediately after payment of any dividend. A dividend of 6% of the value of the stock will be paid in 6 months' time. Thus the dividend paid is $£0.06S_{6/12}$ for each unit of stock held.

The risk-free force of interest is 5% p.a.

1. Using a no arbitrage argument, derive the 9-month forward price of the stock.
2. Calculate the forward price if the dividend was paid in 3-months' time rather than 6-months' time.

Question 11 A one-year forward contract on a non-dividend paying stock is entered into when the underlying stock price is £40 and the one-year risk-free force of interest is 10% p.a. Denote by f_t the value of the *long* forward contract at time t years after the contract is struck.

1. Calculate the forward price and the initial value f_0 of the forward contract.
2. Six months later, the price of the stock is £45 and the 6-month risk-free force of interest is 10% p.a. Calculate the forward price and the value $f_{6/12}$.
3. After a further three months, the price of the stock is £43.50 and the 3-month risk-free force of interest is 8% p.a. Calculate the forward price and the value $f_{9/12}$.

Question 12 A 2-year forward contract on a non-dividend-paying stock is entered into when the underlying stock price is £10. At time 0, the yield on a 2-year zero-coupon bond is 5% p.a. effective.

Consider the following two portfolios.

Portfolio A: long in a 2-year forward contract to deliver one unit of stock for $£K$.

Portfolio B: x units of the 2-year zero-coupon bond and y units of the stock.

1. Use a replication argument to derive a value for the delivery price K.

After one year,

- the stock price is £9.90, and
- the yield on a 1-year zero-coupon bond is 4% p.a. effective.

2. Calculate the no-arbitrage forward price at time 1.
3. Calculate the value at time 1 of a long position in the original 2-year forward contract.
4. Show that Portfolio B at time 1 is equal to the value at time 1 of the original forward contract.

Numerical solutions: Q1.2. 16p; Q1.3. -4 units of A and $+5$ units of B; Q2. Find and demonstrate how: -4 units of A, and $+5$ units of B constitutes an arbitrage strategy; Q3.1. £44.217644; Q3.2. Find and demonstrate how: -1.360544 units of cash, -0.428571 units of A, and $+1$ unit of B constitutes an arbitrage strategy; Q4.1. 81.616107; Q5. Find and demonstrate how: long 1 forward at $K = 102$, -1 unit of stock, and 100 units of cash deposited in the bank at time 0 constitutes an arbitrage strategy; Q6.1. £9 766.93; Q6.2. Find and demonstrate how: short 1 forward at $K = 10\,450$, $+1$ unit of stock, and £10 000 borrowed from the bank at time 0 constitutes an arbitrage strategy;Q9. 1010.05; Q10.1. £97.9445; Q10.2. £97.9445 (i.e. the same as Q10.1); Q11.1. $f_0 = £0.00$ and $K_0 = £44.21$; Q11.2. $f_{6/12} = £2.95$ and $K_{6/12} = £47.31$; Q11.3. $f_{9/12} = £0.17$ and $K_{9/12} = £44.38$; Q12.1. £11.0250; Q12.2. £10.2960; Q12.3. - £0.7010.

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Solutions to Tutorial 8 – Arbitrage and forward contracts

Solution 1 This is a modification of a past professional examination question.

1. The no arbitrage assumption means that it is assumed that an investor is unable to make a risk-free trading profit.

2. In all states of the world at time 1, the payoff from Security B is 80% of the payoff from Security A. Therefore the no-arbitrage price of B must be 80% of the market price of A, i.e. the no-arbitrage price of B is $80\% \times 20\text{p} = 16\text{p}$.

3. As the market price of B is 15p then an investor could short-sell 4 units of Security A and buy 5 units of Security B at time 0. This results in a profit of $+4 \times 20 - 5 \times 15 = 5\text{p}$ at time 0 and no risk of future loss since the net payoff at time 1 is zero.

Solution 2 Note that the payoff on A is 1.25 times the payoff on B, in all future states of the world.

Consider a portfolio that consists of -4 units of A (short-sell A) and $+5$ units of B (buy B).

Setting this portfolio up at time 0 produces an instant profit of $+4 \times 10 - 5 \times 6 = 10$.

At time 1, the payoff on this portfolio is 0 in both states of the world.

That is, we have a sure profit at time 0 of 10 units with no risk of a loss at time 1.

Solution 3 Consider the Portfolio 1 and its replicating portfolio, Portfolio 2:

1. Portfolio 1: hold 1 unit of asset B (i.e. long in B).

   Portfolio 2: deposit x units of cash in the bank, and hold y units of asset A.

		Price at time 1	
Portfolio	Price at time 0	Market falls	Market rises
1	P_B^{fair}	40	55
2	$x + 100y$	$1.05x + 90y$	$1.05x + 125y$

   For replication, we require the payoffs at time 1 to be identical in all states of the world. Hence we have two equations:

$$1.05x + 125y = 55$$
$$1.05x + 90y = 40.$$

   Solving for x and y results in

$$y = 15/35 = 0.428571 \qquad \text{and} \qquad x = 1.360544.$$

   The assumption of no arbitrage means that the value of both portfolios must be the same at time 0. Hence

$$P_B^{\text{fair}} = x + 100y = 44.217644.$$

2. If $P_B < P_B^{\text{fair}}$ then consider again Portfolios 1 and 2.

   Portfolio 1: hold 1 unit of asset B (long in B).

   Portfolio 2: deposit $x = 1.360544$ units of cash in the bank and hold $y = 0.428571$ units of asset A.

   The prices are as in the table below.

Portfolio	Price at time 0	Price at time 1	
		Market falls	Market rises
1	P_B	40	55
2	$P_B^{\text{no arb}}$	40	55

   [As $P_B < P_B^{\text{fair}}$, we pay less for Portfolio 1 than Portfolio 2, but we have identical payoffs in all states of the world. Remember: buy low (cheap), sell high (expensive). So buy Portfolio 1 and (short-)sell Portfolio 2.]

   An arbitrage opportunity is obtained by short-selling Portfolio 2 and buying Portfolio 1. This results in a positive profit at time 0 of $+P_B^{\text{fair}} - P_B > 0$. As the net payoff at time 1 is zero in all states of the world, there is no risk of future loss.

Solution 4

1. $K = S_0 e^{\delta T} = 80e^{0.06\times 4/12} = 81.616107$.

2. The price at time 0 is the result of: (i) the use of a static hedging strategy that does not rely on the statistical model for S_t; and (ii) the principle of no-arbitrage.

Solution 5 The no-arbitrage forward price is

$$K^{\text{no arb}} = S_0 e^{\delta T} = 100e^{0.05\times 0.5} = 102.531512.$$

The proposed forward price of $K = 102$ is therefore not equal to the no-arbitrage forward price and hence an arbitrage opportunity exists.

Buy low, sell high:

- take out a long position in the forward contract (i.e. agree to buy the asset at $K = 102$ in 6 months' time),
- short-sell the portfolio which replicates the no-arbitrage forward price $K^{\text{no arb}} = 102.531512$.

Therefore, at time 0:

- set up a long forward at $K = 102$,
- short-sell one unit of stock at price $S_0 = 100$,
- deposit the $S_0 = 100$ received into a bank account.

At time $T = 6$ months we have cash of $S_0 e^{\delta T} = 100e^{0.05\times 0.5} = £102.531512$. Of this, we use £102 to pay the fixed leg of the forward contract. In return we receive one unit of stock, which cancels out the short position set up at time 0.

Thus we have an arbitrage profit of £0.531512 at time $T = 6$ months.

Solution 6

1. $K^{fair} = 10\,000 \times e^{9/12\times 0.05} - 600 \times e^{6/12\times 0.05} = 10\,382.12 - 615.19 = 9\,766.93.$

2. Since the no-arbitrage price of £9 766.93 is less than the forward price £10 450, an arbitrage profit can be made by buying the no-arbitrage replicating portfolio and shorting (selling) the forward contract.

   An arbitrage strategy: at time 0,

   - short the 9 month forward contract at the forward price £10 450 (i.e. agree to sell the stock for £10 450 in 9 months' time),
   - borrow £10 000 at the risk-free force of interest 5% p.a, and
   - buy 1 unit of the stock at a cost of £10 000.

   The net cost of this strategy at time 0 is zero.

   In 3 months' time, use the dividend of £600 received on the stock to repay part of the bank loan. This means that at time 3 months, the value of the bank loan is $10\,000e^{0.05\times 3/12} - 600$.

   In 9 months' time, the value of the remaining loan is

   $$(10\,000e^{0.05\times 3/12} - 600)e^{0.05\times 6/12} = 9\,766.93,$$

   However at this time we receive £10 450 from selling the stock through the forward contract. We use this cash to repay the loan, leaving a net profit of $10\,450 - 9\,766.93 = 683.07$.

   This gives a certain profit of £683.07 for zero investment at time 0.

Solution 7 The delivery date is in 3 months, so set $T = 3/12$.

Denoting by S_t the GBP/USD exchange rate at time t, we know $S_0 = 1/1.55432$.

The country-specific interest rates expressed as an annual force of interest are $\delta_{\text{UK}} = \ln(1.017) = 0.01686$ and $\delta_{\text{US}} = \ln(1.0075) = 0.00747$.

Hence the arbitrage-free forward price is

$$K = S_0 e^{(\delta_{\text{UK}} - \delta_{\text{US}})T} = \frac{1}{1.55432} e^{(0.01686 - 0.00747)\times 0.25} = 0.64488.$$

Thus the terms of the contract are:

- Delivery date in 3 months' time.
- UK firm pays the counterparty USD10 million at the delivery date.
- Counterparty pays the UK firm GBP6.4488 million at the delivery date.

Solution 8 Consider a long position in the forward contract (Portfolio A) and its replicating portfolio (Portfolio B).

Portfolio A: long in one forward contract (i.e. at time T, pay K units of cash and receive one unit of the index).

Portfolio B: at time 0, hold x units of cash and y units of the index. Between times 0 and T, all received dividends are reinvested in the index.

Now consider Portfolio B in more detail.

- Let $y(t)$ be the number of units of the index held at time t. At $t = 0$, $y(0) = y$. Fix $0 < \Delta t \ll 1$. In the short time interval $[t, t + \Delta t)$, we receive dividends and we use the dividends to buy more units of the index. The increase in the number of units over $[t, t + \Delta t)$ is $y(t + \Delta t) - y(t)$. Let us calculate how many units of the index we can buy with the received dividends.

  The dividends received in the time interval $[t, t + \Delta t)$ are approximately equal in value to

  $$\rho(t)\Delta t \cdot y(t) = gS_t\Delta t \cdot y(t).$$

  This cash amount is used to buy units of the index, which costs S_t per unit. This means that we can buy

  $$\frac{gS_t\Delta t \cdot y(t)}{S_t} = g\Delta t \cdot y(t)$$

  units of the index in the time interval $[t, t + \Delta t)$. Hence the increase in the number of units we hold over $[t, t + \Delta t)$ is

  $$y(t + \Delta t) - y(t) = g\Delta t \cdot y(t) \quad \Rightarrow \quad \frac{y(t + \Delta t) - y(t)}{\Delta t} = g \cdot y(t).$$

  Letting $\Delta t \to 0$, we obtain the differential equation

  $$y'(t) = gy(t).$$

  Solving this subject to the boundary condition $y(0) = y$, we get

  $$y(t) = ye^{gt}.$$

- At time T, the cash investment is worth $xe^{\delta T}$ and the value of the index holdings is

  $$y(T)S_T = ye^{gT}S_T.$$

For B to replicate A, we require $S_T - K = xe^{\delta T} + ye^{gT}S_T$. Equating the coefficients of S_T, we find

$$y = e^{-gT} \qquad \text{and} \qquad x = -Ke^{-\delta T}.$$

The assumption of no arbitrage then implies that Portfolios A and B must have the same value at time 0, i.e. $x + yS_0 = 0$. Substituting for x and y and rearranging, we find $K = S_0e^{(\delta - g)T}$, as required.

Solution 9 Consider a long position in the forward contract (Portfolio A) and its replicating portfolio (Portfolio B).

Portfolio A: long in one forward contract (i.e. at time $T = 2$, pay K units of cash and receive one unit of the index).

Portfolio B: at time 0, hold x units of cash and y units of the index. Between times 0 and T, all received dividends are reinvested in the index.

Denote the dividend yield per unit of index at time t by $g(t)$, with

$$g(t) = \begin{cases} 0.05 & \text{if } 0 \le t < 1 \\ 0.03 & \text{if } 1 \le t < 2. \end{cases}$$

and consider Portfolio B in more detail.

- Denote by $y(t)$ be the number of units of the index held at time t in Portfolio B, with $y(0) = y$. Following the argument of the preceding answer, we find $y'(t) = g(t)y(t)$ and hence
$$y(t) = y \exp\left\{\int_0^t g(u)du\right\}.$$
- At time $T = 2$, the cash investment is worth $xe^{\delta T} = xe^{2\delta}$ and the value of the index holdings is
$$y(T)S_T = y \exp\{0.05 \times (1-0) + 0.03 \times (2-1)\}S_2 = ye^{0.08}S_2.$$

For B to replicate A, we require $S_2 - K = xe^{2\delta} + ye^{0.08}S_2$. Equating the coefficients of S_2, we find
$$y = e^{-0.08} \qquad \text{and} \qquad x = -Ke^{-2\delta}.$$
The principle of no arbitrage then implies that Portfolios A and B must have the same value at time 0, i.e. $x + yS_0 = 0$. Substituting for x and y and rearranging, we find
$$K = S_0\, e^{2\delta - 0.08} = 1\,000\, e^{0.045\times 2 - 0.08} = 1\,010.05.$$

Solution 10

1. Consider the following two portfolios.

   Portfolio A: Enter a forward contract to buy the asset in 9 months for the price of K.

   Value of Portfolio A at time 0 is 0.

   Portfolio B: Hold x units of cash and y units of the stock at time 0. Reinvest any received dividends in the stock.

   Value of Portfolio B at time 0 is $x + yS_0 = x + 100y$.

   Consider Portfolio B at $t = 6/12$. Receive a dividend of $0.06S_{6/12} \times y$. Immediately use the dividend to buy additional units of stock, i.e. purchase $0.06S_{6/12} \times y/S_{6/12} = 0.06y$ units of stock.

   Thus at $t = 6/12$, the total number of units held in the stock is $y + 0.06y = 1.06y$.

   Now consider the value of the portfolios at time $T = 9/12$.

   Portfolio A:
   - Pay K and in return receive one unit of stock worth S_T.
   - Value of A is $S_T - K$.

   Portfolio B:
   - Hold $xe^{\delta T}$ units of cash and $1.06y$ units of stock.
   - Value of B is $xe^{\delta T} + 1.06yS_T$.

   For B to replicate A, we require $S_T - K = xe^{\delta T} + 1.06yS_T$. Equating the coefficients of S_T, we find
   $$y = 1/1.06 \qquad \text{and} \qquad x = -Ke^{-T\delta}.$$
   The principle of no arbitrage then implies that Portfolios A and B must have the same value at time 0, i.e. $x + yS_0 = 0$. Substituting for x and y and rearranging, we find
   $$K = \frac{S_0}{1.06}e^{\delta T} = \frac{100}{1.06}e^{0.05(9/12)} = £97.9445.$$

2. The forward price will be the same if the dividend is payable in 3 months. The *amount* of dividend paid in 3 months' time is $0.06S_{3/12}$ per unit stock holding, so we hold y units of stock at time 3/12, we receive a dividend payment of $0.06yS_{3/12}$. However, we use this payment to buy additional units of stock at time 3/12, which costs $S_{3/12}$ per unit. Thus we purchase $0.06y$ units of stock, exactly as in the previous part of the question. Nothing else changes.

Solution 11

1. Forward price is $K_0 = 40e^{0.10\times 1} = £44.21$.

   Initial value of the long forward contract is $f_0 = 0$.

2. Forward price is $K_{6/12} = 45e^{0.10\times 6/12} = £47.31$.

   Value of the original long forward contract is

   $$f_{6/12} = (K_{6/12} - K_0)e^{-0.10\times 6/12} = £2.95.$$

   Note: the value of the *short* forward contract is $-f_{6/12} = -£2.95$.

3. Forward price is $K_{9/12} = 43.5e^{0.08\times(1-9/12)} = £44.38$.

   Value of the original long forward is

   $$f_{9/12} = (K_{9/12} - K_0)e^{-0.08\times(1-9/12)} = (44.38 - 44.21)e^{-0.08\times 0.25} = £0.17.$$

   Note: the value of the *short* forward contract is $-f_{9/12} = -£0.17$.

Solution 12

1. Denote by S_t the price of the stock at time t and denote by P_t the price of the 2-year ZCB at time t.

   At time 2,

   - Portfolio A has value $S_2 - K$.
   - Portfolio B has value $x + yS_2$

   For replication, we require $S_2 - K = x + yS_2$. Equating the coefficients of S_2, we obtain $y = 1$ and $x = -K$.

   Then no arbitrage implies that values of the portfolios at time 0 are equal. Hence $xP_0 + yS_0 = 0$. Substituting for $x = -K$ and $y = 1$ and using the facts that $P_0 = 1.05^{-2}$ and $S_0 = 10$, we find $K = S_0/P_0 = 1.05^2 \times 10 = 11.025$

2. Using the same argument as above, we find $K_1 = S_1/P_1 = 9.90 \times 1.04 = 10.296$.

3. The value of a long forward contract is

   $$(K_1 - K) \times P_1 = (10.296 - 11.025)/1.04 = -£0.7010$$

4. Recall that portfolio B has $x = -K = -11.025$ units of the bond and $y = 1$ unit of stock. At time 1, this has value

   $$-11.025P_1 + S_1 = -11.025/1.04 + 9.90 = -£0.7010.$$

   This is as it should be. If B replicates A at time 2 and there is no arbitrage then the value of B should track the value of A at all times and not just be equal at time 0.